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Source: *The American Economic Review*, Vol. 84, No. 5 (Dec., 1994), pp. 1141-1156

Published by: American Economic Association

Stable URL: <http://www.jstor.org/stable/2117765>

Accessed: 05-11-2015 23:05 UTC

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The Economic Consequences of Unwed Motherhood: Using Twin Births as a Natural Experiment

By STEPHEN G. BRONARS AND JEFF GROGGER*

We estimate the short-run and life-cycle effects of unplanned children on unwed mothers by comparing unmarried women who first gave birth to twins with unwed mothers who bore singletons. We find large short-term effects of unplanned births on labor-force participation, poverty, and welfare reciprocity among unwed mothers, but not among married mothers. Although most of the adverse economic effects of unplanned motherhood dissipate over time for whites, there are larger and more persistent negative effects on black unwed mothers. (JEL I30, J13)

For a quarter century social scientists have focused attention on the relationship between family structure and poverty and have generally concluded that premarital childbearing exacerbates the problems of both poverty and family instability. It has been argued repeatedly that out-of-wedlock and adolescent childbearing reduces the mother's educational attainment, lowers the probability of her eventual marriage, increases her probability of welfare reciprocity, and decreases family income (see Frank Furstenberg, 1976; James T. Trussell, 1976, 1988; J. J. Card and L. L. Wise, 1978; Sandra Hofferth and Kris Moore, 1979; Charles Murray, 1984; Frank Mott and William Marsiglio, 1985; Furstenberg et al., 1987; Hofferth, 1987; Dawn Upchurch and James McCarthy, 1990). The apparently deleterious effects of early childbearing are central to recent explanations of the growth of an urban underclass, to the increasing number of children born into poverty, and

to the intergenerational transmission of poverty (see Mary Jo Bane, 1986; Bane and David Ellwood, 1986; Irwin Garfinkel and Sara McLanahan, 1986; William J. Wilson and Kathryn Neckerman, 1986; Wilson, 1987; Ellwood, 1988; Gregory Duncan and Saul Hoffman, 1990).

Some economic analyses support the notion that unplanned births may adversely affect a woman's prospects in both the marriage and labor markets. Gary S. Becker (1981) argues that out-of-wedlock children decrease the likelihood of the mother's future marriage by reducing her net resources and raising the cost of searching for a husband. The presence of out-of-wedlock children also raises the value of her nonmarket time and is therefore expected to reduce her labor-force participation and schooling.

Despite the intuition that unplanned premarital childbearing may adversely affect economic outcomes, convincing and credible measurement of the economic consequences of unplanned premarital childbearing has proved to be difficult. First, births are influenced by decisions of the prospective mother. While survey evidence indicates that only about one-sixth of all births to unwed mothers are planned,¹ estimates obtained by treating all out-of-wedlock

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¹Authors' calculations based on figures presented in Trussell (1988).

childbearing as exogenous may nonetheless be inconsistent.

Second, those women who bear children out of wedlock may well be the same women whose economic status would have been lowest in any case. That is, if there are unobservable factors correlated with fertility and the mother's future economic success, then a negative correlation between premarital childbearing and economic success may arise even if unplanned births have no causal influence on economic outcomes.

In this paper, we use an exogenous fertility event, the birth of twins, to estimate the economic effects of unplanned births. Our method, introduced by Mark Rosenzweig and Kenneth Wolpin (1980a,b), compares the economic outcomes of women experiencing premarital twin first births to those experiencing premarital single first births. This approach identifies the consequences of an unplanned birth on women's future fertility and marital decisions, education, labor-force participation, labor earnings, welfare receipt, and poverty status. Under some assumptions, these effects may be used to identify the costs of an unplanned singleton birth.

We use 1970 and 1980 Census micro data to identify twin first births to unmarried women. We compare the socioeconomic outcomes in this treatment sample to those in a randomly selected control sample of unmarried women with single first births. Moreover, the successive cross-section samples of unmarried mothers allow us to follow a cohort of women over time. Our estimates provide insights about life-cycle and cross-cohort changes in the consequences of unplanned births.

Generally, we can reject the hypothesis that unplanned premarital fertility has no adverse economic consequences. We present evidence that the effects of unplanned births differ substantially by race and across cohorts of unwed mothers. Consistent with other recent studies (e.g., Shelly Lundberg and Robert D. Plotnick, 1990; Arline T. Geronimus and Sanders Korenman, 1991), we find that out-of-wedlock births carry consequences considerably smaller than those estimated in earlier studies.

I. Methods and Data

A. Methodology

A research strategy based on the experimental ideal, that is, random assignment of first births, is obviously infeasible. Rather, we exploit as our treatment the random occurrence of twins at the unwed mother's first birth. That is, we compare the outcomes of an unwed mother who bore twins at her first birth with the outcomes of an unwed mother who had a single child, using the difference in their outcomes as our estimate of the effect of unplanned fertility. This is the "twins-first" methodology utilized by Rosenzweig and Wolpin (1980b) to study life-cycle fertility and labor supply in a general sample of mothers.

Assuming momentarily that twinning probabilities are constant over age at first birth, randomly occurring twin births represent truly exogenous events that can be used to estimate the effect of unplanned births on our outcomes of interest, uncontaminated by either simultaneity bias or the influence of heterogeneity in the cost of childbearing. Conditioning on first births assures that the twin outcome is also uncorrelated with preferences for the number of children, since all women desiring any positive number of children must have a first child.² In contrast, as pointed out by Rosenzweig and Wolpin (1980b p. 336), including mothers of twins occurring at any point in the birth order would tend to oversample women who had several births, which in general would result in an oversampling of women who desired larger families. Thus the failure to condition on birth order would contaminate our estimates with unobservable heterogeneity in desired fertility.

The twins-first approach allows us to measure the effect of unplanned fertility using simple statistical methods. In principle, the effect could be estimated as the

²Of course, a relatively small fraction of women desire zero children, and they may react differently to unplanned births.

difference in mean outcomes between the treatment and comparison samples. Twinning probabilities are biologically related to age at first birth, however, which introduces a slight complication. Age at first birth is also known to be correlated with some of our outcome variables, most notably labor-force participation. Thus the failure to control for this factor could result in inconsistent estimates.

To account for this potential problem, we employ the following regression models to estimate the treatment effects:

$$(1) \quad k_i = \alpha_k + \beta_k T_i + \gamma_k A_i + u_{ki}$$

$$i = 1, \dots, N$$

$$(2) \quad Z_{ji} = \alpha_j + \beta_j T_i + \gamma_j A_i + u_{ji}$$

$$j = 1, \dots, J; i = 1, \dots, N$$

where k_i is the number of children born to individual i by the Census date and Z_{ji} is the j th outcome variable for individual i . T_i is a binary twin-birth indicator, and A_i is age at first birth, measured in quarters.³ The independence between T_i and control variables that may also influence k_i and Z_{ji} allows us to subsume these variables into the disturbance terms u_{ki} and u_{ji} . The effect of the unplanned birth on total fertility reported below is the least-squares estimate of β_k , the coefficient of the twin-birth indicator in (1). The effects on other outcome variables are given by the least-squares estimates of β_j , $j = 1, \dots, J$, in (2).⁴

³We checked for possible nonlinearities in the relationship between age at first birth and the outcome variables by generalizing equations (1) and (2) to include quartics in age at first birth. For the most part, the higher-order terms were insignificant; even when they were not, changes in the twin coefficients were generally limited to the third decimal place.

⁴Many of the outcomes we analyze are binary. For these we report generalized least-squares (GLS) estimates to account for heteroscedasticity. Logit estimation in all cases gave nearly identical average derivative estimates. We present the GLS coefficients, however, because they have a natural interpretation as the average change in a probability due to the unplanned birth.

Consider an alternative interpretation of these estimates based on a model in which the number of children affects the unwed mother's socioeconomic outcomes:

$$(3) \quad Z_{ji} = \mu_j + \delta_j k_i + \varepsilon_{ji}$$

$$j = 1, \dots, J; i = 1, \dots, N.$$

If ε_{ji} and k_i are correlated, perhaps because those women with lower economic prospects are more likely to wish to bear more children, the least-squares estimator of δ_j will be biased. In this case the twin-birth indicator T_i can be used as an instrumental variable to estimate δ_j consistently. Note that having twins does not necessarily raise lifetime fertility by one child; a twins-first mother may adjust her future fertility to compensate for the unplanned birth. Thus the estimated birth effects differ across specifications (2) and (3) because β_j/β_k , the instrumental-variables estimate of δ_j , rescales the effects into "per-child" units. These instrumental-variables estimates of the effect of an additional child are also discussed below.⁵

B. Data

The main drawback of the twins-first approach is its reliance on a relatively rare event. Previous researchers have generally analyzed fertility surveys, which provide relatively few twin births and virtually none among the category of unwed mothers. Moreover, information on many of the outcomes of interest to economists is lacking from general fertility surveys.

The Public Use Microdata Samples (PUMS) of the 1970 and 1980 Censuses (Bureau of the Census, 1972, 1983) provide information that allows the researcher to

⁵Age at first birth is also included in these second-stage regressions, because it is known to be correlated with the instrument obtained from the first-stage regression. The independence of T_i and the other determinants of Z_{ji} allows us to subsume into the disturbance terms other regressors which normally might appear on the right-hand side of (3).

match children and their mothers within households and to determine whether children were born as twins. Containing records on millions of individuals, the PUMS data can be used to create sizable twin-birth samples, even for relatively small target population groups such as unwed mothers.

The data we use were drawn from the 1-in-100 State, County-Group, and Neighborhood Characteristic samples from the 5-percent questionnaire of the 1970 Census, and from the A and B samples of the 1980 Census. Together, the 1970 samples constitute a 3-percent sample of the U.S. population, and the 1980 data yield a 6-percent sample. The PUMS data contain a single record for each individual in each household, which includes information on his or her demographic and economic characteristics. The records also contain information about the individual's relationship to the householder and to other persons in his or her household. The PUMS files provide data on several socioeconomic outcomes.

In extracting data on mothers of twins, we first identified individual families within households, then determined whether the first birth of the mother in the family had been to twins. If (i) a woman was living with the same number of her own children as she reported ever having borne, and (ii) her oldest child was age 18 or younger, and (iii) she had a second child with the same age and calendar quarter of birth as the oldest child, then that woman was determined to have had twins first and was included in the sample. We excluded mothers of children over the age of 18 and mothers who were living apart from their children to ensure that we sampled women whose first birth, rather than the birth of the oldest children remaining at home, had been twins. Women without twins who met conditions (i) and (ii) were randomly sampled at rate 0.01 to construct a comparison sample of roughly the same size as the treatment sample.⁶

A total of 3,668 twins-first mothers were located in the 1970 PUMS files, and 5,741 non-twin mothers were sampled as controls. In the 1980 PUMS files, 8,706 twins-first mothers were located, and 12,568 non-twin mothers were included in the control sample. Births were classified as premarital either if the mother reported never having married or if the child's age exceeded the difference between the mother's age and her reported age at first marriage. In the 1970 sample, 289 of the twins-first mothers had their first child out of wedlock, compared to 572 of the mothers in the control sample. Of the twins-first mothers located in 1980, 1,393 had their first child out of wedlock, compared to 2,209 of the control mothers. The sample sizes differ across 1970 and 1980 for two reasons. First, our 1970 Census data set is a 3-percent sample of the United States, and our 1980 data set is a 6-percent sample. In addition, the ratio of unwed to wed births roughly doubled between 1970 and 1980. Thus our sample of unwed twins-first births is roughly four times larger in 1980 than in 1970.

To assess the success of our algorithm in locating twin births, we compare characteristics of our sample with twin-birth characteristics as provided in fertility surveys and vital-statistics publications. First, the probability of a twin-first birth implied by our algorithm is 0.0051 in the 1970 sample and 0.0063 in the 1980 sample. These figures are roughly consistent with the 0.0076 twinning probabilities for first births from the pooled 1965 National Fertility Survey and 1973 National Survey of Family Growth, as reported by Rosenzweig and Wolpin (1980b). Next, our twinning probability for black mothers, 0.007 in both 1970 and 1980, is higher than twinning probabilities for white mothers, which were 0.004 in 1970 and 0.006 in 1980. This is consistent with vital-statistics data tabulated from birth certificates in the United States (U.S. Department of Health and Human Services, 1988 pp. 237–38).

⁶We restrict our samples in the empirical work to include only unwed mothers who first gave birth at age 13 or older. This selection rule excludes 33 unwed mothers (both twin and singleton) in 1970, and 30

unwed mothers in 1980. Although these women had no Census allocation flags and met our other selection criteria, their age at first birth seemed, in many cases, implausibly young.

Finally, regressions of twinning probabilities on the mother's age at first birth indicated that older women are more likely to have twins, consistent with fertility survey data (Rosenzweig and Wolpin, 1980b p. 336). In particular, a regression of a binary twin-birth indicator on the mother's age at first birth (measured in quarters) gave coefficients of 0.002 (SE = 0.001) in 1970 and 0.003 (SE = 0.001) in the 1980 sample.

Table 1 displays means and standard deviations of variables for twin and non-twin unwed mothers in 1970 and 1980. The first row of Table 1 indicates that 40–50 percent of the women in each sample are black, and 40–50 percent are white (Asians and “other” make up the balance). Blacks are overrepresented in these samples of unwed mothers, which is consistent with previous studies (e.g., Wilson, 1987; Trussell, 1988; Duncan and Hoffman, 1990). Average age at first birth, given in the second row of Table 1, is about 3–4 quarters later, on average, for twin mothers compared to our control sample of singleton mothers.

The remaining rows of the table show the means for the number of children born and the other socioeconomic outcome variables for our sample of twin and non-twin mothers in 1970 and 1980. The socioeconomic variables we consider are current marital status, whether the initially unwed mother ever married, years of schooling completed, high-school graduation, labor-force participation, the mother's earnings in the previous year, the total earnings of the mother and her husband (if present) in the previous year, welfare receipt, and poverty status.

We forgo a discussion of differences in means across treatment and control samples in this section; the twin/non-twin differences in Table 1 are identical to the regression results which would be obtained by omitting A_i from equations (1) and (2). We discuss regression results from the less restrictive specification below. Note from Table 1 that the proportion of women ever married in our sample is substantially lower for blacks, and that it falls over time for all women. Mean educational attainment and high-school graduation rates are lower for blacks than whites, but increase over time

for both racial groups. Although labor-force participation rates rise between 1970 and 1980, mean individual and family earnings (in 1979 dollars) fall. The declining mean income in our sample is representative of the experiences shared by many demographic groups over this time period (e.g., Frank Levy, 1987). Welfare reciprocity and poverty rates are substantially higher for blacks than whites. Welfare reciprocity rates grew sharply over time for all unwed mothers, but poverty rates remained fairly stable between 1970 and 1980.

II. Results

A. Unplanned Premarital Childbearing and the Mother's Economic Success

Table 2 presents estimates of the effects of unplanned out-of-wedlock births on a variety of socioeconomic variables for our samples from the 1970 and 1980 Censuses. As noted in Section I, all reported effects are the estimated regression coefficients on a twin-birth dummy variable in regressions that also include age at first birth. We would expect to find modest effects overall if the effect of an unplanned birth diminishes as the child ages and the mother has time to adjust her behavior. We present these overall estimates to facilitate comparison of our results with those of previous authors and to provide a benchmark for our later results.

The first column of Table 2 reports the estimated effects of an unplanned birth for the entire sample in 1970. The second and third columns report separate estimates for the subsamples of whites and blacks. The corresponding effects for the 1980 sample are shown in the last three columns. The standard errors for the 1970 estimates are about twice as large as those in the 1980 sample because of the difference in sample sizes across censuses.

The results in Table 2 show that an unplanned out-of-wedlock birth has a strong positive effect on the number of children in the family. On average, unmarried women with twin-first births had 0.81 more children than unmarried women with single first

TABLE 1—VARIABLE MEANS (STANDARD DEVIATIONS) FOR UNWED MOTHERS

A.						
Variable	1970 Census twin mothers			1970 Census non-twin mothers		
	Overall	White	Black	Overall	White	Black
Sample size	289	123	142	572	306	214
Age at first birth (in quarters)	82.8 (17.2)	84.9 (18.2)	80.0 (15.2)	78.8 (16.1)	80.1 (16.0)	76.0 (13.2)
Number of children	3.16 (1.54)	2.92 (1.07)	3.30 (1.77)	2.42 (1.43)	2.44 (1.35)	2.37 (1.50)
Currently married	0.581	0.772	0.401	0.635	0.755	0.458
Ever married	0.758	0.886	0.634	0.822	0.931	0.659
Years of education	10.89 (2.50)	11.19 (2.54)	10.80 (2.27)	10.70 (2.44)	11.13 (2.16)	10.62 (2.20)
High-school graduate	0.498	0.585	0.444	0.486	0.565	0.430
Labor-force participation	0.405	0.358	0.437	0.430	0.418	0.481
Mother's earnings	3,181 (4,919)	2,804 (4,710)	3,569 (5,138)	2,948 (4,517)	3,054 (4,798)	3,235 (4,436)
Family earnings	10,996 (9,725)	14,297 (10,020)	8,091 (8,076)	11,757 (9,952)	14,291 (10,238)	8,885 (8,900)
Received welfare	0.152	0.024	0.261	0.126	0.065	0.192
Poverty status	0.325	0.203	0.441	0.306	0.193	0.435
B.						
Variable	1980 Census twin mothers			1980 Census non-twin mothers		
	Overall	White	Black	Overall	White	Black
Sample size	1,393	681	607	2,209	1,215	833
Age at first birth (in quarters)	82.3 (15.3)	83.3 (15.4)	80.6 (15.0)	79.7 (14.2)	80.4 (13.5)	77.6 (14.2)
Number of children	2.75 (0.99)	2.65 (0.88)	2.82 (1.03)	2.03 (1.09)	2.00 (0.99)	2.03 (1.17)
Currently married	0.529	0.686	0.325	0.595	0.736	0.390
Ever married	0.687	0.838	0.501	0.728	0.854	0.547
Years of education	11.42 (2.37)	11.56 (2.41)	11.49 (1.97)	11.50 (2.18)	11.44 (2.07)	11.79 (2.02)
High-school graduate	0.650	0.677	0.656	0.645	0.646	0.682
Labor-force participation	0.526	0.518	0.535	0.557	0.514	0.631
Mother's earnings	3,558 (4,925)	3,418 (4,707)	3,830 (5,256)	3,821 (5,009)	3,449 (4,792)	4,522 (5,221)
Family earnings	10,802 (11,304)	13,293 (12,293)	7,977 (9,635)	11,764 (10,923)	13,712 (10,989)	9,298 (9,974)
Received welfare	0.275	0.194	0.387	0.200	0.123	0.301
Poverty status	0.370	0.264	0.488	0.266	0.177	0.375

TABLE 2—EFFECTS OF UNPLANNED BIRTHS ON THE ECONOMIC OUTCOMES OF UNWED MOTHERS

Variable	1970 Census			1980 Census		
	Overall	White	Black	Overall	White	Black
Number of children	0.808 (0.105)	0.530 (0.136)	1.015 (0.174)	0.750 (0.036)	0.681 (0.045)	0.833 (0.059)
Currently married	−0.047 (0.035)	0.029 (0.046)	−0.040 (0.054)	−0.069 (0.017)	−0.050 (0.022)	−0.066 (0.026)
Ever married	−0.058 (0.029)	−0.041 (0.030)	−0.008 (0.052)	−0.038 (0.016)	−0.009 (0.017)	−0.038 (0.027)
Years of education	0.113 (0.178)	−0.064 (0.241)	0.096 (0.241)	−0.149 (0.076)	0.024 (0.104)	−0.400 (0.104)
High-school graduate ^a	−0.062 (0.064)	−0.195 (0.103)	0.037 (0.084)	−0.075 (0.033)	−0.054 (0.050)	−0.108 (0.047)
Labor-force participation	−0.036 (0.036)	−0.076 (0.052)	−0.057 (0.054)	−0.035 (0.017)	0.001 (0.024)	−0.105 (0.026)
Mother's earnings	66 (335)	−512 (505)	151 (513)	−388 (169)	−155 (227)	−911 (275)
Family earnings	−983 (715)	−381 (1,088)	−806 (939)	−1,192 (378)	−650 (550)	−1,589 (523)
Received welfare	0.027 (0.025)	−0.041 (0.022)	0.071 (0.046)	0.080 (0.014)	0.073 (0.017)	0.092 (0.025)
Poverty status	0.023 (0.034)	0.011 (0.043)	0.019 (0.054)	0.111 (0.016)	0.098 (0.019)	0.124 (0.026)
Sample size:	861	429	356	3,602	1,896	1,440

Note: Numbers in parentheses are standard errors.

^aSample restricted to women whose first birth occurred before age 17½. The numbers of observations in each subsample are: 963 overall, 439 white, and 471 black in 1980; 253 overall, 91 white, and 129 black in 1970.

births in 1970, and 0.75 more children in 1980. There are substantial racial differences in these effects: black women reduce future childbearing by less than white women on average in response to an out-of-wedlock twin-first birth.

The results from the 1980 sample also indicate that unplanned out-of-wedlock births had a significant negative effect on marriage propensities, labor-force participation, and earnings for black women. In the treatment sample, blacks' labor-force participation rates were 10.5 percentage points lower, individual earnings were \$900 less (about 22 percent of the overall sample mean), and family earnings were \$1,600 less (about 18 percent of the overall sample mean), than in the comparison sample in

1980. The effects on poverty and welfare are somewhat larger in 1980 than in 1970, and larger for black mothers than white mothers. For example, among black unmarried mothers in 1980, the unplanned child increased welfare reciprocity rates by 9.2 percentage points, and poverty rates by 12.4 percentage points.

An important issue in the study of unwed childbearing is its effect on the school completion of unwed mothers. We use two different measures of education: years of schooling completed and high-school graduation, defined as having completed at least 12 years of school. To estimate the effects on high-school graduation, we restrict the sample to women whose first child was born before they normally would have graduated

TABLE 3—EFFECTS OF UNPLANNED BIRTHS ON THE ECONOMIC OUTCOMES OF MARRIED MOTHERS

Variable	1970 Census			1980 Census		
	Overall	White	Black	Overall	White	Black
Number of children	0.638 (0.027)	0.617 (0.028)	0.794 (0.121)	0.676 (0.014)	0.668 (0.015)	0.765 (0.059)
Currently married	-0.001 (0.006)	-0.001 (0.006)	0.010 (0.033)	-0.004 (0.005)	0.001 (0.005)	0.016 (0.026)
Years of education	-0.029 (0.054)	-0.024 (0.054)	0.168 (0.195)	-0.053 (0.038)	-0.026 (0.038)	0.004 (0.133)
High-school graduate ^a	-0.015 (0.037)	-0.040 (0.041)	0.061 (0.119)	0.034 (0.035)	-0.014 (0.039)	-0.098 (0.098)
Labor-force participation	-0.023 (0.011)	-0.021 (0.012)	-0.057 (0.039)	-0.010 (0.008)	-0.011 (0.008)	-0.041 (0.026)
Mother's earnings	-124 (103)	-147 (111)	-147 (384)	-52 (88)	-100 (93)	83 (330)
Family earnings	-242 (267)	-178 (291)	-125 (773)	-166 (206)	-59 (222)	897 (662)
Received welfare	0.010 (0.003)	0.007 (0.003)	0.012 (0.022)	0.013 (0.003)	0.009 (0.003)	-0.003 (0.019)
Poverty status	0.032 (0.007)	0.023 (0.006)	0.089 (0.034)	0.040 (0.005)	0.029 (0.004)	0.062 (0.024)
Sample size:	8,554	7,387	663	17,672	15,428	1,282

Note: Numbers in parentheses are standard errors.

^aSample restricted to women whose first birth occurred before age $17\frac{3}{4}$. The number of observations in each subsample are: 908 overall, 724 white, and 110 black in 1980; 535 overall, 410 white, and 68 black in 1970.

high school, including only women whose first birth occurred before the age of $17\frac{3}{4}$.⁷ In the 1980 sample, the overall effect of an unplanned additional child was to reduce the probability of high-school completion for black women by 10.8 percentage points. For the entire sample of black unwed mothers, an unplanned birth reduces mean years of schooling by 0.4 years. In contrast, the unplanned child has no significant effect on the educational attainment of white mothers.

We also used the instrumental-variables (IV) approach discussed above to estimate the effect of an extra child on the various

socioeconomic outcomes. These IV estimates are the ratio of the twin effect on the given socioeconomic outcome (β_i) to the effect of twins on the number of children (β_k), after adjustment for age at first birth.⁸ For example, the IV estimate for years of education in 1980 is -0.199, with a standard error of 0.101; the ratio of relevant coefficients from Table 2 is also -0.199 ($= -0.149/0.750$). Other estimates may be constructed similarly.

It is instructive to contrast the results in Table 2 with the consequences of an unplanned birth among married women. Table 3 displays estimated unplanned fertility effects for women who were married at the

⁷The youngest high-school graduates are those who were born in the summer and started first grade within three months of turning six years old. At the end of the 12th grade, these students are between 17 and 18 years old, that is, under age $17\frac{3}{4}$.

⁸The IV estimates are exactly the ratios of the regression coefficients in (1) and (2). This fact was first noted in a different context by Joshua Angrist (1990).

time of their first birth for the 1970 and 1980 samples. In general, the coefficients are small and insignificantly different from zero. Because marriage is a choice variable, however, we do not view the differences in the effects of unplanned births by marital status as predictions of the effectiveness of policy measures intended to promote marital over unwed childbearing.

We also constructed a sample of unmarried mothers who had twins at their second birth, and a randomly chosen comparison sample of women who had at least two children out of wedlock but did not have twins second. We used this sample to analyze the effect of an unplanned child on a mother never married before her second birth, estimating regression models which included the twin indicator, age at first birth, and age at second birth. Sample sizes were smaller; only 118 twins-second births to unmarried women were found in the 1970 Census, while 444 were found in the 1980 Census. As a result, none of the unplanned birth effects estimated from the 1970 sample was significant.

Several of the effects estimated with the 1980 data were significant, however, and are interesting to summarize briefly to provide a sense of how the effects of an unplanned birth may vary with birth order. Twins at the second birth increased the unwed mother's number of children by 0.781 (SE = 0.058) on average. This figure differed little by race; the number was 0.711 (0.090) for white mothers, and 0.826 (0.077) for black mothers. In contrast to twins at the first birth, having twins second had no effect on the probability of marriage, either at the census date or at any time since the second child's birth. Only 46 women had a twin-second birth out of wedlock before turning $17\frac{3}{4}$ years old. This number was too small to assess the effects of such a birth on high-school completion. Unwed mothers who had twins second substantially reduced their labor-force participation in response to the unplanned birth. The effect was -0.123 (0.033) overall, -0.130 (0.058) for whites, and -0.138 (0.042) for blacks. The extra child at the second point in the birth order apparently raised the value of the mother's

time in home production by a greater amount than an unplanned child at the first birth. Estimates of the effects on family earnings were rather large, but generally imprecisely measured.

Perhaps the sharpest contrast between the effects of twins first and twins second was in welfare reciprocity. The estimated coefficients in the twins-second sample were all less than 0.03, and never larger than their standard errors. Welfare participation was much higher in the twins-second sample than in the twins-first sample: 39 percent of all unwed mothers in the twins-second sample received welfare, compared with 23 percent in the twins-first sample. Of course many unwed mothers would be eligible for public support after the birth of their first child; the diminishing marginal effect of an unplanned child on welfare reciprocity therefore may stem from greater previous use of welfare among the twins-second mothers.

B. *The Effects of Unplanned Births over the Life Cycle*

Since money and particularly time costs of childbearing may vary substantially with the child's age, it is interesting to measure the effects of unplanned births at different points in the life cycle. We exploit successive cross-sections of Census data to examine the same cohort of mothers at two points in time. The cohort of women whose first child was born between 1966 and 1969 is observed first in 1970, when the children were 0–3 years old, then again in 1980, when the children were ages 10–13. Our ability to follow a single cohort of mothers over time should yield better estimates of changes in the effects of an unplanned birth over the life cycle than estimates by age group from a single cross-section. In fact, we find that the confounding of age and cohort effects in a single cross section sometimes gives misleading impressions of the change in the effect of an unplanned child over time.

In Table 4, one follows the cohort of mothers whose first childbirth occurred between 1966 and 1969 by moving diagonally

TABLE 4—UNPLANNED BIRTH EFFECTS OVER THE LIFE CYCLE AND ACROSS COHORTS OF UNWED MOTHERS

Variable	Census year	Age of first child, all mothers		Age of first child, white mothers		Age of first child, black mothers	
		0–3	10–13	0–3	10–13	0–3	10–13
Number of children	1970	0.852 (0.075)		0.851 (0.113)		0.849 (0.104)	
	1980	0.970 (0.031)	0.732 (0.080)	0.927 (0.040)	0.678 (0.094)	1.026 (0.051)	0.812 (0.139)
Currently married	1970	–0.131 (0.069)		–0.069 (0.106)		–0.134 (0.085)	
	1980	–0.063 (0.034)	–0.092 (0.035)	–0.047 (0.047)	–0.010 (0.041)	–0.031 (0.043)	–0.142 (0.060)
Ever married	1970	–0.152 (0.068)		–0.098 (0.098)		–0.146 (0.091)	
	1980	–0.037 (0.034)	–0.035 (0.026)	–0.016 (0.046)	0.025 (0.025)	0.0005 (0.045)	–0.073 (0.052)
Years of education	1970	0.197 (0.325)		0.540 (0.464)		–0.288 (0.367)	
	1980	–0.153 (0.130)	–0.089 (0.167)	–0.067 (0.181)	0.177 (0.214)	–0.234 (0.170)	–0.516 (0.259)
Labor-force participation	1970	–0.114 (0.065)		–0.086 (0.094)		–0.206 (0.092)	
	1980	–0.123 (0.033)	–0.013 (0.036)	–0.139 (0.045)	–0.037 (0.050)	–0.155 (0.051)	–0.007 (0.057)
Mother's earnings	1970	–432 (480)		–335 (815)		–1,113 (640)	
	1980	–598 (214)	–326 (397)	–731 (296)	–709 (548)	–658 (345)	–131 (657)
Family earnings	1970	298 (1,275)		4,183 (2,171)		–3,029 (1,455)	
	1980	–872 (574)	–1,406 (841)	–1,130 (882)	–197 (1,186)	–72 (725)	–2,878 (1,229)
Received welfare	1970	0.065 (0.051)		–0.009 (0.056)		0.133 (0.079)	
	1980	0.112 (0.032)	0.087 (0.027)	0.088 (0.041)	0.057 (0.028)	0.136 (0.053)	0.130 (0.056)
Poverty status	1970	0.028 (0.067)		–0.016 (0.094)		0.123 (0.097)	
	1980	0.147 (0.033)	0.127 (0.031)	0.143 (0.044)	0.068 (0.035)	0.154 (0.052)	0.168 (0.058)
Sample size:	1970	220		88		111	
	1980	885	760	460	430	353	275

Note: Numbers in parentheses are standard errors.

from the cell for 0–3-year-olds in 1970 to the cell for 10–13-year-olds in 1980. We also present estimates of the effects of unplanned children on unmarried women whose first birth occurred between 1976 and 1979, whose children were 0–3 years old in 1980. Comparing the effects on cohorts of

women with young children ten years apart provides some evidence on changes in the circumstances facing unmarried women with young unplanned children during the 1970's.

The first two rows of Table 4 give the effects of twin births on the number of children. Among mothers with children 0–3

years old at the Census date, the average time since birth is roughly 1.5 years. For both childbearing cohorts, the estimates can be interpreted as indicating that the effect of a twin birth is to increase the number of children in the mother's family by about 1 for the first 18 months after the birth.

Over time, of course, the mother may adjust her subsequent fertility in response to the surprise birth. By 1980, the number of additional children in the families of twins mothers whose first birth occurred in the late 1960's was about 0.73. The figure is significantly different from unity, indicating that, on average, women of this cohort did adjust their fertility. The amount of adjustment over the ten-year period is about the same as that reported by Rosenzweig and Wolpin (1980b table 2).

Fertility adjustment appears to differ by race. Among white unwed mothers, the number of additional children fell from 0.85 to 0.68 in the ten years following the twin birth, and both figures differ significantly from unity at conventional levels. For black unwed mothers, however, the number of additional children remains relatively constant, and neither estimate differs significantly from 1. It would appear that black unwed mothers adjust their fertility by less than their white counterparts in response to an unplanned birth.

Fertility adjustment also differs by marital status at first birth. In regressions not reported here, we found that married mothers of twins of the late-1960's childbearing cohort had 0.866 (SE = 0.022) more children, and those of the late-1970's cohort had 0.893 (0.013) more children on average. By 1980, however, married twins-first mothers of the early cohort had on average only 0.584 (0.031) children more than married non-twin mothers. There were some racial differences among married mothers: white twin mothers of the late-1960's cohort had 0.832 (0.024) additional children in 1970 and 0.562 (0.031) in 1980, while for blacks the respective figures were 0.886 (0.123) and 0.707 (0.124). Although the estimates for married blacks show a relatively small decline, the 1980 figure is significantly less than 1. In general, then, women married at their first

birth show a substantially greater reduction in additional fertility following a twin birth than do initially unmarried mothers.

The next rows of the table report the effect of an unplanned child on the probability that the mother who was unwed at childbirth was married at the Census date. Members of the late-1960's cohort who had unplanned births were, on average, 13 percentage points less likely to be married while their children were young, and the differential had narrowed only to 9 percentage points by the time their children were pre-teenagers. These women's unplanned births carried a substantial and at least marginally significant marriage penalty.

This effect also appears to vary by race. At both points in time, the coefficients for white mothers are insignificantly different from zero. For black mothers, however, the estimates are numerically large and at least marginally significant, and they do not fall over time. Although the racial differences are not statistically significant, they suggest that unwed black mothers of the late-1960's cohort suffered a greater marriage penalty as a result of an unplanned birth than did white mothers.

In contrast, women of the late-1970's cohort who bore twins experienced little if any reduction in the probability of marriage while their children were young. The overall estimate is half the size of that for the earlier cohort at the same point in the life cycle and is only marginally significant. Neither of the race-specific estimates is larger than its standard error.

Consider next the effect of unplanned births on the probability that a mother who was unmarried at the time of her first birth married at any time between the birth and the Census date. In general, the results are similar to those above. There is a sizable negative effect in the short run for the earlier cohort, and essentially none for the later cohort. As one possible explanation for this change, we note that the number of births per 1,000 unmarried women increased from 26.4 in 1970 to 29.4 in 1980 (Bureau of the Census, 1986 p. 62). That is, the relative supply of unmarried women with children increased. The decrease in the

marriage penalty suffered by unwed women with unplanned children may therefore result from the increasing prevalence of unwed women with children in the marriage market.

Alternatively, the changes in marriage penalties may be a result of changes in the cost of obtaining an abortion following *Roe v. Wade*. Before the 1973 Supreme Court decision, abortions were equally unavailable to women regardless of the stability of the relationship that resulted in the premarital pregnancy. After the liberalization of abortion laws, women for whom an out-of-wedlock child would be most costly could choose to terminate the pregnancy. Presumably, the cost of an unplanned child would be lower for women whose pregnancy stemmed from a relationship that was more likely to result in marriage.⁹

The final point to be taken from these estimates is methodological. Following the earlier cohort of black mothers over time, we see a roughly constant differential in current marriage rates across treatment and control samples (13.4 percentage points in 1970 and 14.2 percentage points in 1980). Looking at black women at two different points of the child-age distribution in 1980, however, gives the impression that the marriage-rate differential increased from 3.1 to 14.2 percentage points over a ten-year period. These substantially different estimates of marriage-rate differentials caution against using age-group differences to approximate changes in effects of unplanned fertility over the life cycle.

We next examine the effects of unplanned births on years of education. The effects, in general, are imprecisely measured. The only marginally significant effect

is for black women in the early childbearing cohort in 1980, who had 0.52 years less education as a result of an unplanned birth.

The estimated effects on labor-force participation are more striking. Table 4 shows that the presence of an unplanned young child substantially reduces the labor-force participation of the unmarried mother in the short run, with little impact after ten years. In the first three years of the child's life, the probability that the mother works or looks for work is reduced by about 11–12 percentage points, a number that is constant across cohorts and somewhat larger for black women. By the time the child is 10–13 years old, however, the unplanned birth has no effect on the mother's participation. This is consistent with the interpretation that an unplanned child increases the mother's marginal rate of substitution between consumption and leisure but that the magnitude of this increase diminishes as the child ages.

The effects of the unplanned child on the mother's earnings are presented next. For the full sample, the estimates are statistically significant only for the later childbearing cohort in 1980. Disaggregating by race, we see that the earnings of black mothers of the early cohort in 1970 also fell in response to the surprise birth.

The effects on family earnings, defined as the sum of the mother's earnings plus those of her spouse, if present, range widely and are imprecisely estimated for the overall sample. The same is true for the white mothers, for whom unplanned children have no statistically significant effect on family earnings. This contrasts sharply with the results for black mothers from the early cohort. An unplanned child reduced black unwed mothers' family earnings by an average of over \$3,000 in 1970, or one-third of average black family earnings. Moreover, these effects were persistent. In 1980, black women who had twins out of wedlock more than a decade earlier still had family earnings one-third lower than their counterparts. Unplanned childbearing imposed substantial and significant costs on black unwed mothers of the early cohort. For the late-1970's cohort, however, the initial earnings effect was essentially zero. This is prob-

⁹It is also possible that these differences could result from relatively greater use of elective abortion to terminate multiple pregnancies. This would generally require the mother to be aware of her multiple pregnancy during the first trimester, since later abortions carry substantially greater risks. Although ultrasound technology can identify multiple gestations during the first trimester, obstetric use of ultrasound was still quite rare in the 1970's (F. H. Dalicho, 1989).

ably the most substantial cross-cohort change in Table 4. Whereas black unwed mothers who had unexpected children in the 1960's experienced large and significant decreases in family resources, women in the same position ten years later experienced none.

We consider some possible explanations for this substantial change. Decreases in family earnings can result from decreases in own earnings, decreases in the husband's earnings, or falling marriage rates. The effects on own earnings and participation rates in Table 4 cannot provide a complete explanation of the effect on family earnings. Participation rates and earnings fall for both cohorts but remain low for the early cohort only while their children are young. In contrast, marriage rates were significantly and persistently reduced by the presence of unplanned children for the earlier cohort, but not for the later cohort. It therefore would appear that the changes in the way unplanned children affect subsequent marriage have had an important bearing on the family earnings of unwed mothers.

We next examine the effect of unplanned children on the welfare reciprocity of unwed mothers. The overall estimates indicate that unplanned births raised the probability of welfare participation for both cohorts, with the later cohort experiencing the largest initial effects. The increase among the early cohort was persistent; welfare reciprocity rates remained higher for mothers of young unplanned children a decade after their first children were born.

There are substantial differences in welfare effects by race, however. Among the early cohort, white unwed mothers with unplanned children were no more likely to receive welfare than their counterparts. By the time their children were pre-teenagers, they were actually 5.7 percentage points more likely to receive welfare on average. Across cohorts of white unwed mothers, however, there was a substantial increase in welfare participation. For the later cohort, an unplanned child increased the likelihood of welfare receipt by a significant 8.8 percentage points. In contrast, the effect of unplanned children on the welfare reciprocity of black unmarried mothers was larger

and persistent for the earlier cohort, but it did not vary across cohorts. In other words, the entire increase across cohorts in the effect of unplanned children on welfare receipt stems from the increase among white mothers.

The welfare effects described above are generally mirrored in the results on poverty rates. The overall estimates show that an unplanned birth raises the probability of poverty for both cohorts, with the effect on the later cohort being substantially larger. Disaggregating by race uncovers several important facts. As in the case of welfare reciprocity, there are substantial racial differences in the effects of unplanned childbearing on poverty for the early cohort of mothers in 1970. An unplanned birth has no effect on the poverty rate for white unwed mothers who recently gave birth. By 1980, however, an unplanned child increased the poverty rate by 14.3 percentage points for white unwed mothers and 15.4 percent for black mothers who recently gave birth. Unplanned childbearing effects are persistent; the effect of an unplanned child on poverty status grew between 1970 and 1980 for both whites and blacks of the early childbearing cohort.

C. Intergenerational Consequences of Unplanned Premarital Childbearing

The results presented so far indicate that unplanned out-of-wedlock childbearing causes substantial negative effects on the mother's economic well-being. For white mothers, these effects are relatively short-lived, although there is a fair amount of persistence among black mothers. The magnitudes of these effects, however, particularly on the family earnings of unwed black mothers, open the possibility that the child's schooling level may be reduced during the period over which the mother experiences adverse outcomes. In this case, the negative effects of unplanned premarital fertility might be transmitted to the children. Therefore, even short-term damage to the mother's well-being could have very long-run adverse social consequences.

The twins data can be used to study one aspect of the question of intergenerational

TABLE 5—EDUCATIONAL PROGRESS OF CHILDREN OF UNWED MOTHERS

Variable	1970 Census			1980 Census		
	Overall	White	Black	Overall	White	Black
Constant	-6.30 (0.326)	-6.95 (0.432)	-5.81 (0.448)	-6.21 (0.095)	-6.43 (0.145)	-6.03 (0.126)
Age	0.919 (0.025)	0.952 (0.033)	0.911 (0.033)	0.958 (0.008)	0.966 (0.011)	0.959 (0.010)
Twin indicator	0.045 (0.153)	-0.125 (0.220)	0.161 (0.196)	0.024 (0.043)	0.082 (0.064)	-0.076 (0.057)
R^2 :	0.775	0.799	0.830	0.896	0.822	0.922
Sample size:	415	220	163	1,879	987	753

Note: Numbers in parentheses are standard errors.

transmission. In particular, we can estimate the effect of unplanned premarital births on the educational progress of the children themselves. We thus provide estimates of the extent to which unplanned out-of-wedlock childbearing results in decreased levels of this particularly important form of human-capital investment in the children of unwed mothers.¹⁰

Our approach to this question exploits the simple notion that, after age six, a child progressing through school at the usual rate should complete an additional year of schooling every year, at least until the child reaches the mandatory schooling limit. In other words, in a sample of children ages 7–16, a regression of years of completed education on age should have an intercept of -6 and a slope of 1. To test for the effects of unplanned premarital childbearing, which may result in decreased time or financial inputs into the child's education, we estimate this model, adding the twin indicator to allow for the effects of unplanned births. We therefore compare the schooling progress of a singleton child with the progress of one child of a twin pair, where the choice between twins is made at random. To account for the fact that school entry requirements generally cause children born later in the year to enter school at a

greater age than those born earlier in the year (Angrist and Alan Krueger, 1991), the model also contains a full set of quarter-of-birth indicator variables.¹¹

The results are presented in Table 5. As expected, for all groups, the constant is close to -6, and the slope is close to 1. The twin coefficients, however, are small numbers with large standard errors. We also estimated models that included an interaction of the twin indicator with the age variable, to test whether unplanned children's rate of progress differed from their singleton classmates. The twin dummy and interaction terms were insignificant in those models as well, both individually and jointly. Thus the test provides no evidence that the deleterious effects experienced by the mothers of unplanned children are transmitted to their children.

III. Conclusions

This paper examined the outcomes of a natural experiment, the birth of twins, to determine the socioeconomic consequences of unplanned childbearing by unwed mothers. After controlling for birth order and age at first birth, twin births are truly exogenous with respect to unobserved differences

¹⁰Rosenzweig and Wolpin (1980a) used twin differences to study the child quantity–quality trade-off in a sample of Indian children.

¹¹The inclusion of the mother's age at first birth did not substantially affect the results.

in opportunity costs and preferences for childbearing.

Our estimates indicate that the effects of unplanned births differ in many cases by race and across cohorts of unwed mothers. While nearly all unwed mothers experience substantial short-run decreases in labor-force participation after an unplanned birth, only black women of our early childbearing cohort experienced significant decreases in the probability of eventual marriage or family earnings. For these women, unplanned births had sizable and persistent negative effects.

Across cohorts, we observe a substantial decrease in the negative effects of unplanned out-of-wedlock births. Whether this change stems from decreased stigma associated with increased levels of unwed motherhood or from the increased availability of abortions following the Supreme Court's 1973 decision in *Roe v. Wade* is a question we cannot address with our data. Given the implications for policy, however, this is an important question for future work.

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